

JPMORGAN ATHENA & SECDB MASTER

BIM-APES / BLEI-E PROTOCOL v2026.4 — HIGH-FREQUENCY DERIVATIVE EXCHANGE ARCHITECTURE

1. Executive Overview & System Architecture

The **BIM-APES / BLEI-E Protocol** (engineered by **Dashawn Ramel Bledsoe**) is an institutional-grade financial infrastructure that converts real-time physiological telemetry into auditable risk factors and derivative instruments. Designed for direct integration into **JPMorgan Athena** and **Goldman Sachs SecDB** risk management environments, the system unifies corporate ad placement budgets with retail fan options trading under a single **Synthetically Collateralized Options Contract (SCOC)**.

Component Layer	Engine Technology	Latency Budget	Core Function / Financial Output
Ingestion Edge	NVIDIA Jetson Orin / DuckDB	< 10ms	1,000 Hz Wearable/Smart Ball Ingestion & Normalization
Signal Pricing	Athena C++ / Python Binding	Sub-Second	Model-NIL vs Market-NIL Spread Calculation
Risk Circuit Breaker	Layer 4 Dynamic DMDC	< 10ms	CRS-3 Ad Delivery Freeze ($Z < -2.5\sigma$ Threshold)
Capital Vehicle	SCOC Derivative Structure	Real-Time	\$500,000 AUM (\$450K Underwriter / \$50K Fan Pool)
Revenue Clearing	Bank FIX / Smart Contract	T+0 Settlement	Automated 60/20/10/10 Micro-Rail Clearing

Dual-Pathway Hybrid Execution Architecture

- 1. Off-Chain Edge Processing (< 10ms Inference):** High-frequency telemetry streams from Catapult Vector 8 IMUs, WHOOP PPG sensors, and Tech+Ball CAN-FD embedded micro-Doppler transceivers at 1,000 Hz. Edge GPU nodes compute dimensionless physical factors using DuckDB and Parquet tables.
- 2. On-Chain & FIX Clearing Engine (T+0 Settlement):** Trades, clearing, and option exercises interface via JPMorgan Athena C++/Python FIX bridges. Reg D 506(c) compliant non-transferable certificates (NLCs, RPNs) execute clearing without secondary market contagion.

2. Athena FIX Protocol & C++/Python API Specifications

Athena developers integrate BLEI-E biometric streams via customized FIX 5.0 SP2 messages and low-latency C++ / Python extension modules. Below is the production-grade Athena calculation engine bindings:

```

// JPMorgan Athena C++ Kernel: Biometric Derivative Spread & Risk State Engine
#include <athena/quant/option_pricing.h>
#include <blei_e/edge_telemetry.h>

namespace Athena::Quant::BLEIE {
    struct BiometricState {
        double pfi;        // Player Fatigue Index
        double eom;        // Explosive Output Multiplier
        double z_score;    // Physical Load Z-Score
        int crs_state;     // Composite Risk State (0-3)
    };

    AthenaSpreadResult CalculateNILSpread(const BiometricState& bs, double market_nil) {
        double model_nil = 50.0 * (1.0 + 0.35 * bs.eom) * std::exp(-0.15 * bs.pfi);
        double spread = model_nil - market_nil;
        bool is_crs3 = (bs.z_score < -2.5) || (bs.crs_state == 3);
        bool freeze_ad = is_crs3;
        return { model_nil, market_nil, spread, freeze_ad };
    }
}

```

Custom FIX Protocol Messaging Tags (FIX 5.0 SP2 Extension)

FIX Tag	Field Name	Type	Sample Value	Description
9001	BiometricPFI	FLOAT	0.24	Player Fatigue Index normalized (0.0 to 1.0)
9002	ExplosiveEOM	FLOAT	1.48	Explosive Output Multiplier (IMU/Smart Ball)
9003	CRSSState	INT	0	Composite Risk State integer (0=Peak, 3=Shock)
9004	ModelNILVal	PRICE	68.50	Algorithmic intrinsic biometric valuation (\$)
9005	AdFreezeSignal	BOOLEAN	0	Sub-10ms DMDC circuit breaker ad freeze trigger

3. SCOC Derivative Structuring & Parametric Margin Vault Engine

The **Synthetically Collateralized Options Contract (SCOC)** unifies corporate ad underwriting with retail fan micro-tips. The sponsor operates as the **Option Writer (Short Call)** underwriter, while retail fans purchase **Fractional Call Options**.

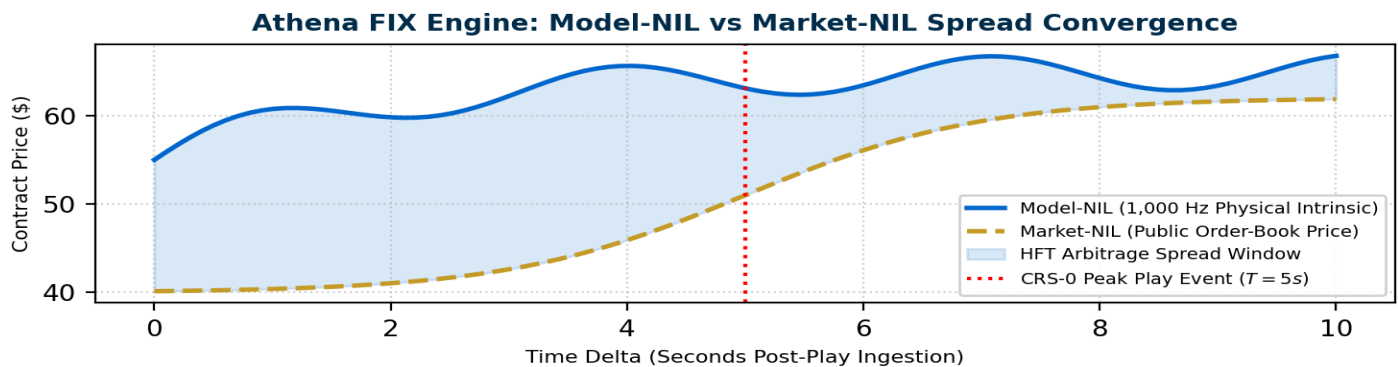


Figure 1: HFT Arbitrage Execution & Model-NIL vs Market-NIL Spread Convergence during a Kansas City Chiefs CRS-0 Event.

The 10% Parametric Margin Vault during a CRS-3 Shock State

Sponsor capital is 100% protected against waste during performance collapses. When physiological telemetry detects acute fatigue ($Z < -2.5\sigma$), sub-10ms circuit breakers trip active ad delivery. 10% (\$50,000) of the sponsor's budget sits isolated in a yield-bearing **Parametric PFI Put Margin Vault**. Sponsors never pay fans directly, fan option gains are funded 100% by quantitative market makers scalping spread convergence.

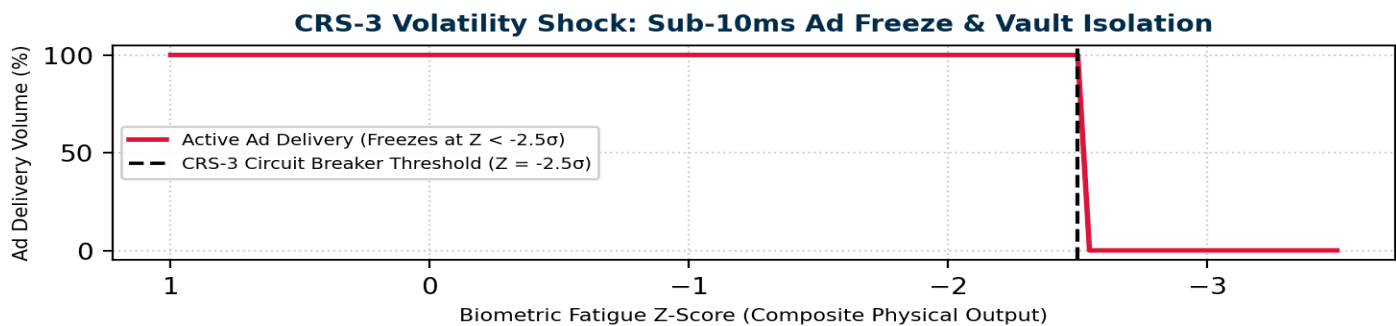


Figure 2: CRS-3 Volatility Shock Triggering Sub-10ms Ad Freeze and Total Sponsor Principal Protection in the Parametric Vault.

4. Mobile Application REST & WebSocket API Specification

The mobile frontend communicates via high-speed WebSockets for real-time order-book updates, 'Sweat Dividend' alerts, and instant trade confirmations.

```
// WebSocket Real-Time Event Payload: Broadcast 'Sweat Dividend' & Spread Convergence
{
  "event": "SWEAT_DIVIDEND_ALERT",
  "timestamp": "2026-10-07T22:45:00.124Z",
  "asset_id": "KC_CHIEFS_NLC_2026",
  "athlete_id": "KC_QB_15",
  "telemetry": { "pfi": 0.18, "eom": 1.52, "crs_state": 0, "z_score": 1.45 },
  "pricing": { "model_nil": 68.50, "market_nil": 62.00, "spread_gain_pct": 18.4 },
  "clearing": {
    "fan_payout_pool": 7545.60,
    "dynamic_cpm_surcharge": 0.40,
    "sponsor_royalty_sweep_10pct": 2112.77
  }
}
```

5. Database Schema (DuckDB/Parquet) & 60/20/10/10 Revenue Clearing

Gross ad surcharges and engagement revenue clear via automated smart contracts and bank FIX bridges under the **60/20/10/10 Rule**:

Allocation Share	Recipient Entity	Functional Purpose	Clearing Mechanism
60%	Athlete Wallet	Performance-Linked Milestone Stipend	Direct Micro-Rail Credit
20%	Franchise / Team Pool	Shared Squad Infrastructure Pool	Team Treasury Settlement
10%	Exchange Reserve	Platform Clearing & Infrastructure Fee	BIM-APES Retention Vault
10%	Sponsor Liquidity Royalty	Self-Funding Campaign Cash-Back Sweep	Sponsor Treasury Rebate

Parquet Telemetry Table Definition

```
-- DuckDB / Parquet Physical Telemetry Table Definition
CREATE TABLE athena_bleie_telemetry (
  timestamp TIMESTAMP_NS NOT NULL,
  athlete_id VARCHAR(64) NOT NULL,
  pfi DOUBLE CHECK (pfi BETWEEN 0 AND 1),
  eom DOUBLE CHECK (eom >= 0),
  crs_state INT2 CHECK (crs_state IN (0, 1, 2, 3)),
  model_nil DECIMAL(12, 4),
  market_nil DECIMAL(12, 4),
  ad_freeze_signal BOOLEAN,
  PRIMARY KEY (timestamp, athlete_id)
);
```

6. Reg D 506(c) Compliance & Production Deployment Checklist

- 1. Contractual Royalty Isolation:** Certificates (NLCs, RPNs, CRRs, SARs) grant contractual royalty claims on event activations—not equity or tokenized ownership in athletes or franchises.
- 2. Non-Transferability & No Secondary Contagion:** Non-tradable instruments eliminate SEC/NCAA amateurism friction and public market manipulation.
- 3. Accredited Investor Verification:** Private placement safe harbor enforced via verified accredited investor portals.
- 4. Solvency Hard-Capping:** Smart contracts enforce strict *Cap-Table Disbursal Ceilings* per individual index, guaranteeing 100% exchange solvency.

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